NEUROANATOMICAL MARKERS OF SENSITIVITY TO SOCIAL ISOLATION: AMYGDALA SUBSTRUCTURE IN ADOLESCENTS

Do individual differences in amygdala subregion volumes predict psychological responses to acute social isolation in adolescent humans?

AUTHORS

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01 BACKGROUND

Social Connection & Loneliness

- Social connection is an evolutionarily conserved 'innate need' (Matthews & Tye, 2019)
- Loneliness is an adaptive drive to maintain social contact and avoid aversive consequences of social isolation
- Amygdala key in social deficit neural circuits

Evidence

- *Rodents:* isolation = cellular changes in amygdala (Lavenda-Grosberg et al., 2022; Zhang et al., 2022)
- *Humans:* ↑ loneliness → ↓ amygdala activation & ↓ grey matter (Lieberz et al., 2021; Duzel et al., 2019)

Amygdala Subregions

- Amygdala not unitary – discrete subnuclei (Wilensky et al., 2006)
- Distinct functions: aversive processing (Michely et al., 2020), social stress connectivity (Chang & Yu, 2018), adolescent restructuring (Campbell et al., 2021)

PARTICIPANTS

- N = 35 adolescents with complete data (from 42 recruited)
- *Age:* 16–19 years
- *Recruitment:* schools & local advertisements (Cambridge, UK)
- *Exclusion criteria:* neurological conditions, mental health disorders, high chronic loneliness

02 METHODS

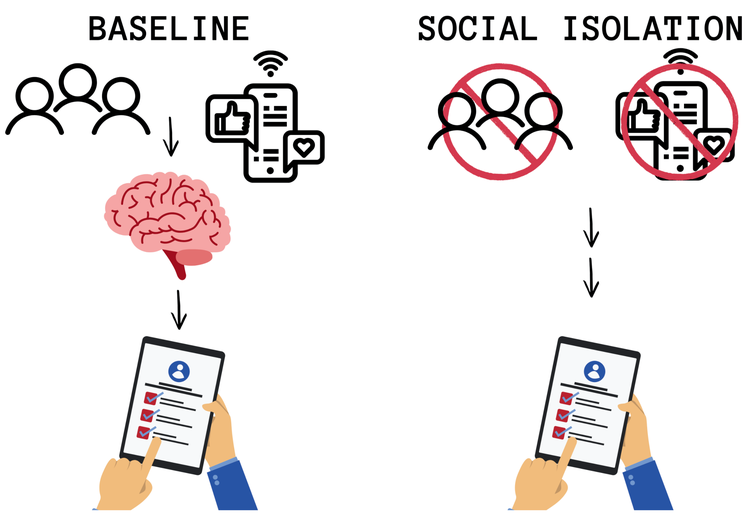
PROCEDURE

Study Design

- *3 sessions:* baseline + 2 counterbalanced isolation sessions
- *Baseline:* brain scan, questionnaires, behavioural tasks
- *Isolation sessions:* 3.5–4 hours controlled lab isolation
  - *One session:* digital social interaction access (data not included here)

Psychological Measures

- *Collected:* at baseline & post-isolation
- *Measured:* state anxiety (STAI), loneliness, social craving, boredom, positive/negative mood (PANAS)



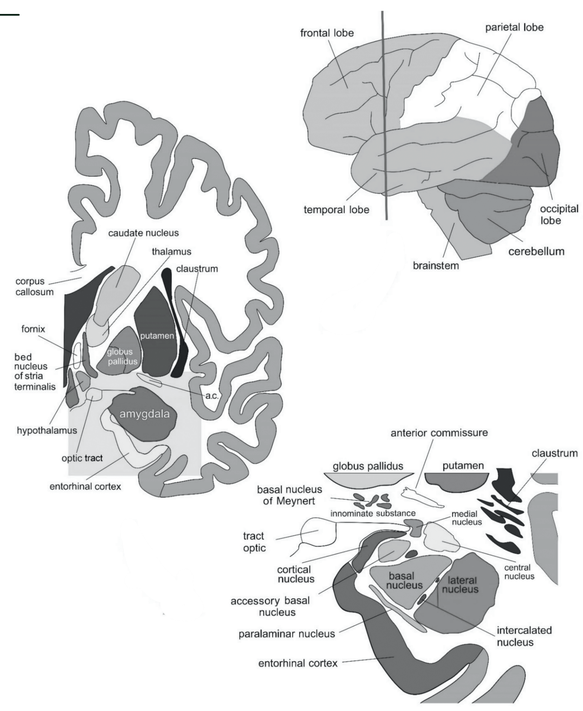
MAGNETIC RESONANCE IMAGING

MRI Acquisition

- 7-Tesla Siemens Terra scanner (Wolfson Brain Imaging Centre, Cambridge, UK)
- *T1-weighted MP2RAGE sequence:* 224 sagittal slices, 0.7mm isotropic voxels, *FOV:* 224mm

Amygdala Segmentation

- *FreeSurfer 7.3:* 9 nuclei per hemisphere
- *Nuclei:* lateral, basal, accessory basal, anterior amygdaloid area, central, medial, cortical, corticoamygdaloid transition, paralaminar
- *Extracted:* subregion volumes + eTIV (control variable)



ANALYSES

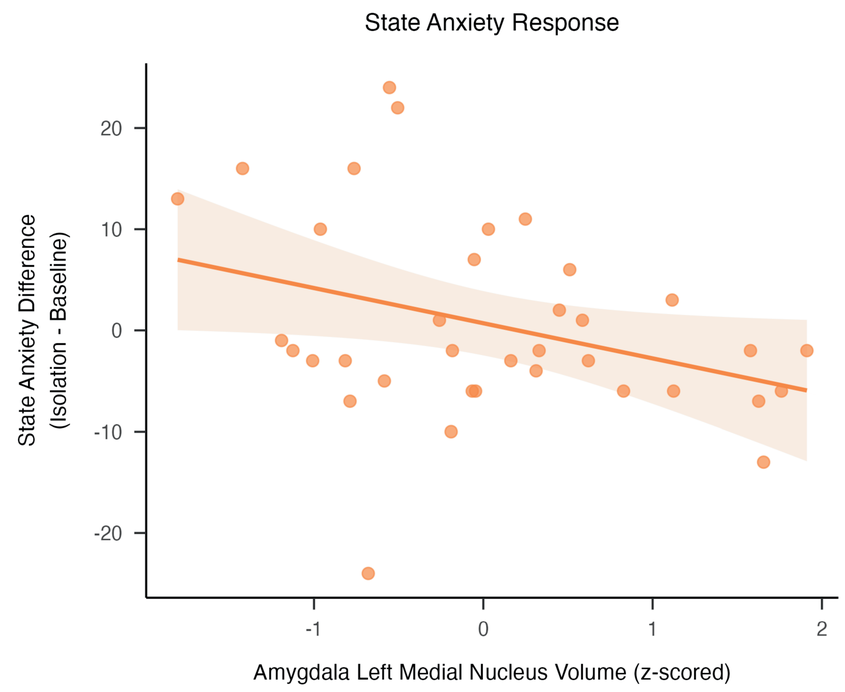
- LASSO regression with 10-fold cross-validation
- *Predictors:* 18 amygdala subregion volumes (9/hemisphere) + eTIV (all standardised)
- *Outcomes:* differences in psychological state (baseline → post-isolation)
- *Lambda:* optimized via cross-validation to minimize prediction error

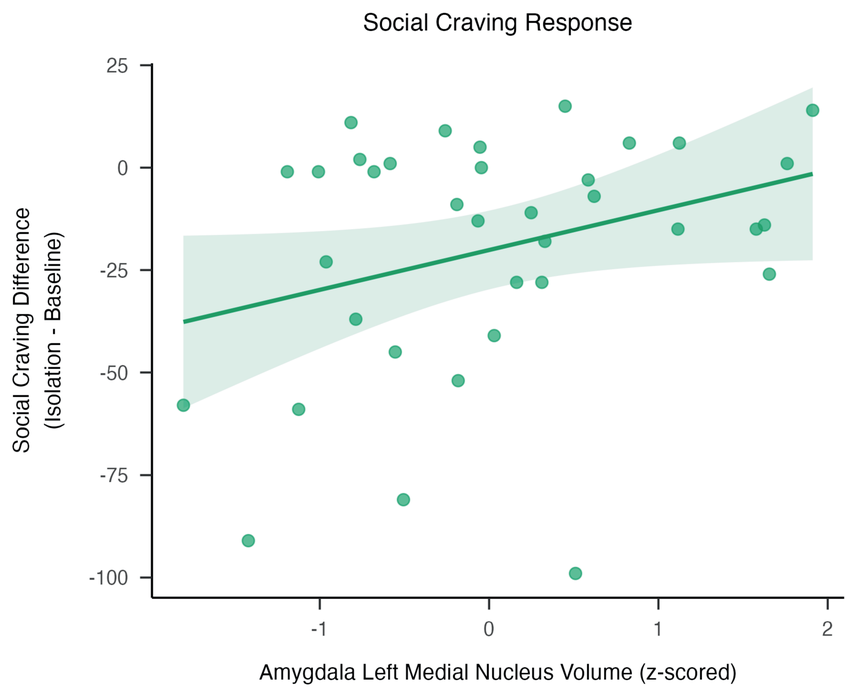
Validation

- *Stability analysis:* 1000 bootstrap iterations
- *Inclusion criteria:* regions selected in initial LASSO AND >70% bootstrap samples

03 RESULTS

- **Left medial nucleus** predicted psychological responses to isolation
  - *State anxiety:* negative prediction ($\beta = -2.01$, $R^2 = 0.14$, 72.8% stability)
  - *Social craving:* positive prediction ($\beta = 6.09$, $R^2 = 0.25$, 74.4% stability)
- No stable predictors for loneliness, boredom, positive/negative mood
- Effects lateralized to left hemisphere only
- eTIV not selected for any outcome





04 DISCUSSION

- Left medial nucleus = neuroanatomical predictor of isolation sensitivity
  - *Structural dissociation:* ↑ volumes → ↓ state anxiety AND ↑ social craving post-isolation

Prior research has documented both **threat hypervigilance** (Towner et al., 2024) and **heightened social motivation** (Tomova et al., 2020) as group-level responses to social isolation. However, the balance between ***social approach*** and ***threat avoidance*** after isolation might be related to individual differences in amygdala morphometry.

These preliminary findings have implications for understanding individual differences in vulnerability to isolation-related psychopathology.

REFERENCES

Campbell et al., (2021). Developmental cognitive neuroscience.
Chang & Yu, (2018). Neurobiology of stress.
Duzel et al., (2019). Scientific reports.
Lavenda-Grosberg et al., (2022). Molecular psychiatry.
Lieberz et al., (2021). Advanced science.
Matthews & Tye, (2019). Annals of the NY academy of sciences.
Michely et al., (2020). Cognitive neuroscience and neuroimaging.
Tomova et al., (2020). Nature neuroscience.
Towner et al., (2024). Royal society open science.
Wilensky et al., (2006). The journal of neuroscience.
Zhang et al., (2022). Molecular psychiatry.

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